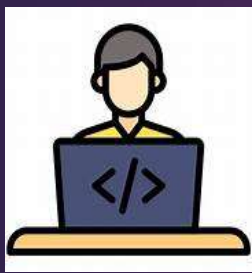


End-to-End DevOps CI/CD Pipeline using Jenkins, Docker & SonarQube



Developer
will write
code



Code will
get into
the github



Code will get
into CI sever
using github
repo



For code
quality
analysis



Dockerfile → **build** → Image
Image → **run** → Container
Container runs the app
(access via IP + port).



Steps:

- ✓ AWS EC2 setup
- ✓ Jenkins installation & configuration
- ✓ Tool integration (JDK, NodeJS, SonarQube)
- ✓ CI pipeline creation
- ✓ Docker image build
- ✓ DockerHub push
- ✓ Container deployment
- ✓ Basic security scanning

Step 1: Launch an EC2 Instance with t2.large and 25GB Volume

The screenshot displays the AWS Management Console interface for the 'us-east-1' region. The left-hand navigation pane shows the 'EC2' section expanded, with 'Instances' selected. The main content area shows a list of instances with one instance, 'Docker', listed. The instance is in a 'Running' state, has an 'Instance ID' of 'i-04e54c1633059bc5c', and is of type 't2.large'. Below the list, the details for the 'Docker' instance are shown, including its 'Instance ID', 'IPv4 address', 'Instance state', 'Public IPv4 address', 'Private IPv4 addresses', and 'Public DNS'.

Instances (1/1)

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4
Docker	i-04e54c1633059bc5c	Running	t2.large	2/2 checks passed	View alarms	us-east-1d	ec2-54-162

i-04e54c1633059bc5c (Docker)

Instance summary

Instance ID: i-04e54c1633059bc5c

IPv4 address: 54.162.27.247 | open address

Instance state: Running

Public IPv4 address: ec2-54-162-27-247.compute-1.amazonaws.com | open address

Private IPv4 addresses: 172.31.43.90

Public DNS: ec2-54-162-27-247.compute-1.amazonaws.com | open address



Step 2: Install Jenkins, Git, Docker, & Trivy

Step 3: Install the following Jenkins Plugins

Sonar Scanner

NodeJs

OWASP

Dependency Check

Docker Pipeline

Eclipse Temurin Installer Version

Pipeline Stage View

Step 4: Configure all the plugins into Jenkins

Trivy Installation:

wget

https://github.com/aquasecurity/trivy/releases/download/v0.18.3/trivy_0.18.3_Linux-64bit.tar.gz

tar xvf trivy_0.18.3_Linux-64bit.tar.gz

sudo mv trivy /usr/local/bin/

vim .bashrc

export PATH=\$PATH:/usr/local/bin/
source .bashrc

Jenkins Installation:

```
amazon-linux-extras install java-openjdk11 -y sudo wget -O  
/etc/yum.repos.d/jenkins.repo  
https://pkg.jenkins.io/redhatstable/jenkins.repo sudo rpm --import  
https://pkg.jenkins.io/redhat-stable/jenkins.io-2023.key yum install  
jenkins -y systemctl start Jenkins
```

GIT & Docker Installations: yum install git docker -y systemctl start
docker

```
chmod 777 ///var/run/docker.sock
```

Setup Sonar using Docker:

```
docker run -d --name sonar -p 9000:9000 sonarqube:lts-community
```

Login to the sonar dashboard with the following and credentials

username: admin

password: admin

Create one Dummy Project and Generate a Security Token in
Sonarqube Dashboard Configure Plugins to Jenkins

Configure Plugins to Jenkins

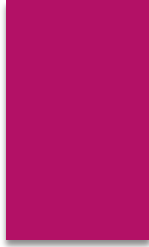
Go to Jenkins Dashboard → Manage Jenkins → Credentials → Add Secret Text with ID “sonar”

Go To Manage Jenkins → System → Add Sonar server with the name of “mysonar” → Save

Click on Add SonarQube Server → Name: mysonar → Click on Install automatically with default version

Configure JDK,NodeJs, & DP-Check

Manage Jenkins → Tools → Add JDK → Name: jdk17 → Install Automatically → Select “Install from adoptium.net” → Select Version as “jdk-17.0.8.1+1”



Next!!

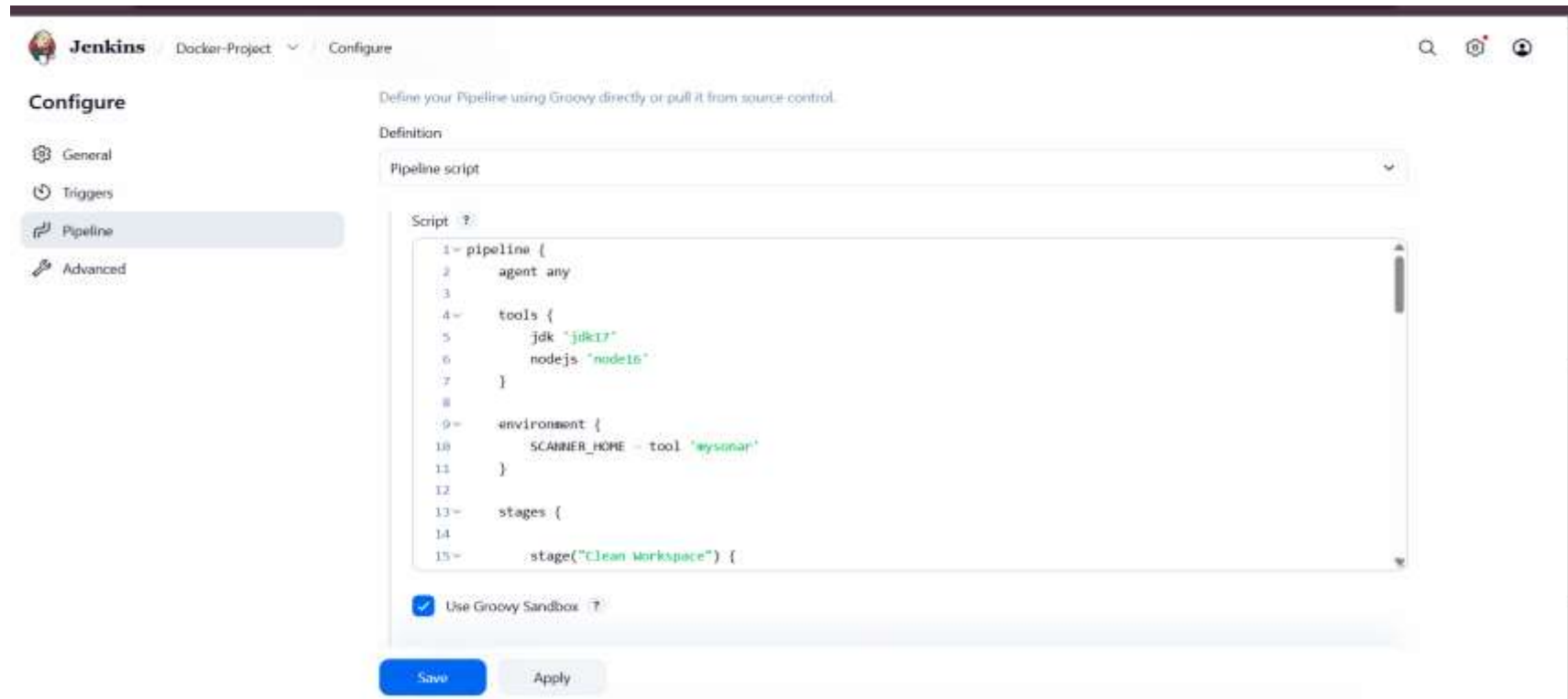
Add NodeJS → Name: node16 → Version:
NodeJS 16.2.0

Next!!

Add Dependency-Check → DP-Check →
Install Automatically → Install from
github.com → Version: 6.5.1

Save

Step 5: Write pipeline script and build



The screenshot shows the Jenkins web interface for configuring a pipeline. The top navigation bar includes the Jenkins logo, the project name 'Docker-Project', and the 'Configure' link. On the left, a sidebar lists configuration options: 'General', 'Triggers', 'Pipeline' (which is selected and highlighted), and 'Advanced'. The main content area is titled 'Configure' and contains the instruction 'Define your Pipeline using Groovy directly or pull it from source control.' Below this, a 'Definition' dropdown menu is set to 'Pipeline script'. A 'Script' section with a help icon (?) contains a text area with the following Groovy pipeline script:

```
1= pipeline {
2   agent any
3
4   tools {
5     jdk "jdk17"
6     nodejs "node16"
7   }
8
9   environment {
10    SCANNER_HOME = tool "qysonar"
11  }
12
13  stages {
14    stage("Clean Workspace") {
```

Below the script text area, there is a checkbox labeled 'Use Groovy Sandbox' which is checked. At the bottom of the configuration area are two buttons: 'Save' (in blue) and 'Apply' (in grey).



Jenkins

Docker-Project

Full Stage View



Docker-Project - Stage View

		Declarative: Tool Install	Clean Workspace	Checkout Code	SonarQube Analysis	Install Dependencies	OWASP Scan	Trivy FS Scan	Build Docker Image	Deploy Container
Average stage times:		176ms	308ms	1s	23s	19s	1min 15s	806ms	2min 5s	755ms
#3	Mar 03 15:36	176ms	308ms	1s	23s	19s	1min 15s	806ms	2min 5s	755ms
	No Changes									

Projects

Docker-Project - Jenkins

React App

Not secure 54.162.27.247:9000/projects

You're running a version of SonarQube that is no longer active. Please upgrade to an active version immediately. [Learn More](#)

sonarqube

Projects

Issues

Rules

Quality Profiles

Quality Gates

Administration

Search for projects...

A

My Favorites

All

Filters

Quality Gate

Passed1

Failed0

Reliability (Bugs)

A rating0

B rating0

C rating1

D rating0

E rating0

Security (Vulnerabilities)

A rating1

B rating0

C rating0

D rating0

E rating0

Security Review (Security Hotspots)

Search by project name or key

Create Project

2 project(s)

Perspective: Overall Status

Sort by: Name

☆ sample

Project's Main Branch is not analyzed yet [Configure analysis](#)

☆ zomato

Passed

Last analysis: 13 minutes ago

Bugs1C

Vulnerabilities0A

Hotspots Reviewed0.0%E

Code Smells0A

Coverage0.0%

Duplications0.0%

Lines1.3kCSS, Jav...

2 of 2 shown

Embedded database should be used for evaluation purposes only

The embedded database will not scale, it will not support upgrading to newer versions of SonarQube, and there is no support for migrating your data out of it into a different database engine.

SonarQube™ technology is powered by SonarSource SA

Community Edition - v9.9.0 (build 100196) [NO LONGER ACTIVE](#) - [LGPL v3](#) - [Community](#) - [Documentation](#) - [Plugins](#) - [Web API](#)

33°C Sunny

Search

ENG IN

58%

15:42 03-03-2026

